HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed manufacturing operations and processes for a semiconductor client, identifying bottlenecks and proposing optimizations to improve productivity.
- Developed and enhanced capacity and headcount models for production operations, incorporating historical data and future demand forecasts.
- Designed and implemented production standards across multiple manufacturing cells, focusing on efficiency and waste reduction.
- Mapped inventory flow and optimized pull systems within and between manufacturing cells, reducing work-in-progress inventory by **15%**.
- Engineered control systems to minimize production issues and reduce costs, integrating real-time monitoring data from shop floor equipment.
- Applied Visual Factory concepts to production lines, creating real-time status displays that improved team awareness and response times.
- Containerized key Java Spring Boot microservices using **Docker** and deployed them to an **AWS EKS** cluster, enabling more reliable and scalable deployments.

SKILLS

Manufacturing Operations – Production machinery, Process optimization, Capacity planning, Headcount modeling, SEMI-E10, SEMI-E79, Inventory flow, Visual Factory concepts

Core Engineering – Process analysis, Productivity improvement, Waste reduction, Control systems design, Production standards, Manufacturing layouts

Programming Languages – Java, Python, SQL

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Agile, Scrum, Jira, Confluence, Dynatrace, Splunk

PROJECTS

Semiconductor Fab Process Efficiency Analysis

- Analyzed complex wafer manufacturing processes for a semiconductor client, identifying key drivers of cycle time variance and areas for improvement.
- Developed a simulation model in Python to test the impact of different scheduling algorithms on overall manufacturing efficiency, projecting a potential **12%** throughput increase.
- Collaborated with operations teams to document existing workflows and propose new process flows that reduced material handling steps by **20%**.

Automated Production Reporting Dashboard

- Built a Python-based ETL pipeline to aggregate data from multiple shop-floor systems and production databases.
- Developed interactive dashboards using **Power BI** to visualize key performance indicators, including cycle time, yield, and equipment uptime for manufacturing operations.
- Implemented automated report generation and distribution via email, reducing manual reporting effort by **8 hours** per week.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)